Using Deep Learning to Optimize Energy Consumption in Data Centers & Supercomputers

Hassan Hankir, Computer Science , h.hankir@aui.ma

Supervisor: Dr. Nasser Assem

8 May, 2026

جامعة الأخوين
AL AKHAWAYN UNIVERSITY
School of Science and Engineering

SPRING'26 CAPSTONE EXHIBITION DAY

Introduction

Modern GPU data centers account for a growing share of global electricity consumption. Traditional schedulers like **Least Loaded (LL)** and **Energy-Aware Ready (EAR)** dispatch jobs on myopic rules — they cannot learn from the environment or anticipate price changes. This project implements a **PPO (Proximal Policy Optimization)** reinforcement learning agent in a custom 8-server GPU cluster driven by real-world electricity pricing, with sleep/wake mechanics and a price-aware task deferral mechanism.

Problem Statement

What: GPU data centers face dual challenges: unpredictable demand + time-varying electricity tariffs. Naive dispatch during peak pricing wastes significant cost.

Why it matters: Data centers consumed an immense amount of energy every year. Intelligent scheduling reduces cost and carbon footprint without sacrificing service quality.

Functional & Non Functional Requirements

Functional: Schedule or defer GPU jobs across 8 servers; enforce SLA (max 8 defers); integrate real electricity pricing; server sleep/wake mechanics.

Non-Functional: Statistically significant cost reduction ($p<0.05$) vs. baselines; reproducible across 5 seeds; modular and scalable to 16/32 servers.

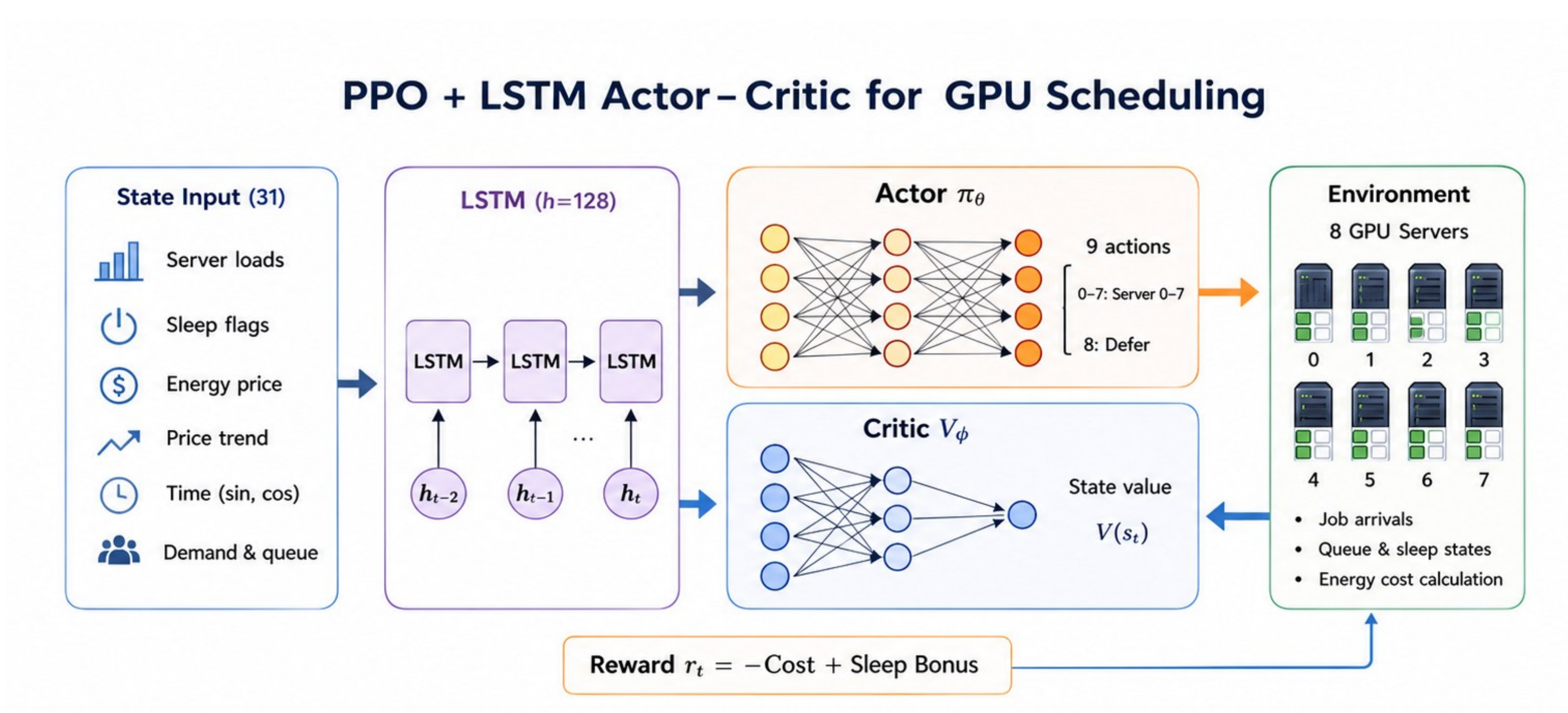
System Design

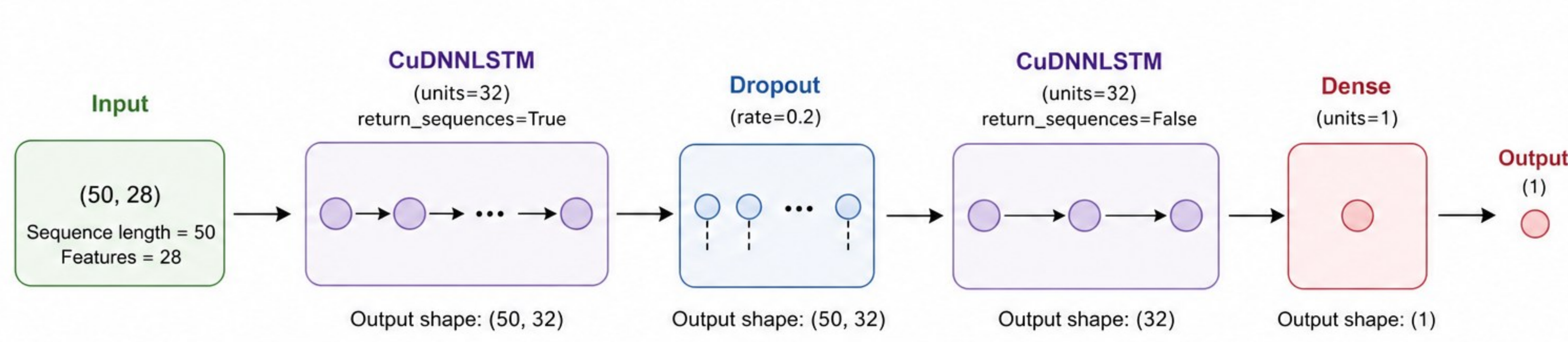
RL Environment: 8 servers, capacity 10 jobs. State dim=31. Actions: 8 dispatch + 1 defer. Poisson arrivals ($\lambda=2.0$).

PPO Agent: LSTM ActorCritic (hidden=128), clipped objective ($\varepsilon=0.2$), cosine LR decay, 2000 episodes/seed × 5 seeds.

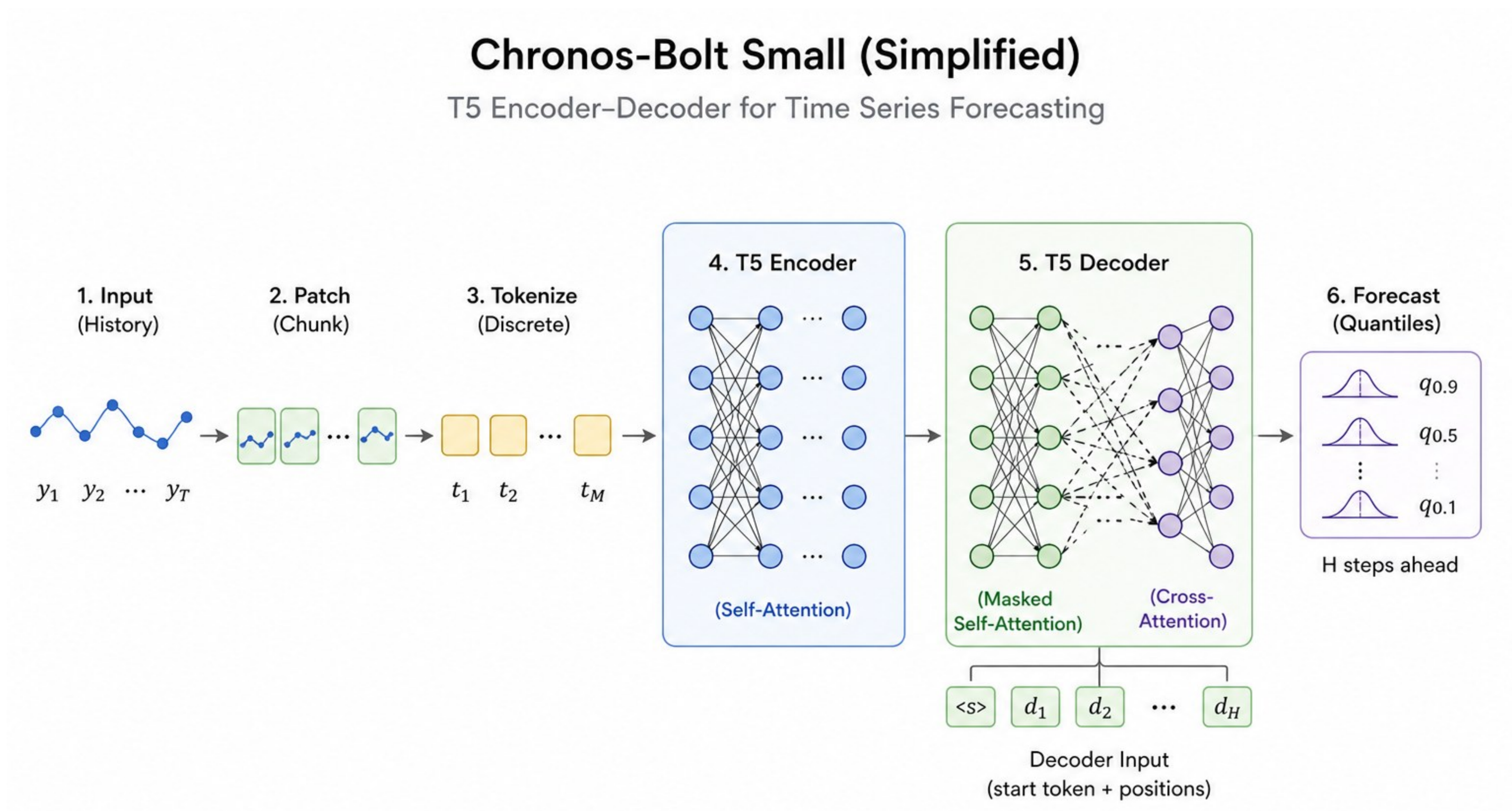
Deferral: Queue up to 20 jobs at price>0.55; $5×10^{-6}$ holding cost/step; SLA force-dispatch after 8 defers.

Supporting Models: 2-layer LSTM (32 units) for load prediction; Chronos-Bolt (48M, zero-shot) for demand forecasting.





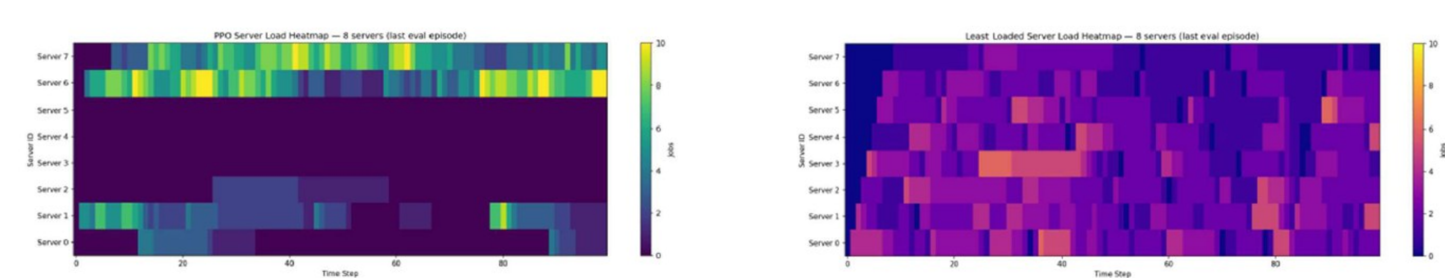
Implementation



Results & Evaluation

31.2%	33.5%
Cost reduction vs. EAR	Cost reduction vs. Least Loaded

- PPO beats both baselines on all 5 seeds
- Agent defers jobs at high price (>$0.08/kWh), resumes when cheap, behavior unreachable by myopic heuristics
- Server heatmaps: PPO concentrates load, keeps idle servers asleep, reduces wake-up penalties
- Chronos-Bolt: accurate 48-step probabilistic forecasts with zero fine-tuning



Conclusion

The PPO agent successfully learns to minimize energy costs in a realistic, price-volatile GPU cluster simulation. Its learned deferral behavior is qualitatively beyond any myopic heuristic.

The multi-seed, paired-test evaluation provides robust statistical guarantees, and the LSTM + Chronos components deliver a scalable, data-driven framework for sustainable GPU cloud scheduling.

AI Use

Code: AI assistants accelerated boilerplate; all code reviewed by author. **Pre-trained Models:** Chronos-Bolt (48M) used zero-shot for demand forecasting.

Challenges & Future work

Challenge: Balancing cost minimization with SLA compliance required iterative reward tuning.

Future: Scale to 16/32 servers; add carbon intensity; integrate Chronos into PPO state; zero-shot policy transfer to EU/US energy data; Kubernetes prototype; conference paper submission.

References

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Acknowledgments

We gratefully acknowledge the support of our **Ecole Polytechnique** and **UC Berkeley**, whose expertise, equipment, and collaboration were essential for the development of this project. We also thank **Al Akhawayn University**, as well as friends and family.

Avenue Hassan II, Ifrane 53000, Morocco.